

this gave the Tomcat a famous and coveted “look-down, shoot-down” capability. Its radar could track up to 24 enemy aircraft simultaneously and lock on to six of them as prime targets for its Phoenix missiles. In addition, onboard computers, fed with data from reconnaissance aircraft and satellites, gave the F-14 access to a wide range of ancillary information on activity across the whole battlefield.

New generation of fighters

The F-14 and F-15 (see pages 320–21) were modern weapons systems of astonishing complexity, and, as such, extremely expensive. The next of the US’s modern fighters, the Lockheed Martin F-16, was a response to the demand for a lighter, cheaper aircraft that could be afforded in greater numbers and would have an emphasis on maneuverability for air-to-air combat. Although it ended up as a complex machine with multirole capability, the F-16 emerged as the most agile of close-combat fighters, setting a standard for aerial maneuver that has not yet been surpassed.

Throughout aviation history, there has been a tradeoff between stability and maneuverability – the more unstable the aircraft, the more agile it will be. The F-16’s designers created an aircraft that was inherently unstable, made flyable by sophisticated electronics. With its fly-by-wire controls, a computer linked to the aircraft’s control surfaces ensured that the right adjustments would be made at every instant to keep the aircraft under control. The limit on the F-16’s high-speed twist-and-turn is human – the amount of g-force that a pilot can withstand without blacking out.

By the time the US Navy’s F/A-18 Hornet went into production in 1987, the rest of the world was struggling to keep up. One reason was cost. Even for the US, aircraft such as the F-14 and F-15 were phenomenally expensive, as was their armament – a Phoenix missile cost around \$500,000. International cooperation just managed to keep Western Europe in the game. A consortium of British, Italian, and German companies created the Panavia Tornado in the 1970s, and another joint European effort generated the Eurofighter for the third millennium. In France, Dassault produced the Mirage 2000 and the Rafale. However, these aircraft were essentially ancillary to the US air forces, representing primarily an effort to stay in touch with advanced technology and keep aerospace industries active.

The Soviet Union was the country that produced the aircraft the American fighters were likely to meet in combat. The Soviets were prepared to invest the resources to keep up with the Americans, but they had a tradition of producing simpler, more robust aircraft that gave them a potential battlefield superiority in

INSIDE A MODERN FIGHTER COCKPIT

AIRCRAFT DESIGNERS HAVE BEEN well aware that the complexity of modern fighters could overload their pilots, however well trained. The modern “glass” cockpit, which first appeared in the F-18 Hornet and has since been retrofitted on other fighters, is designed to provide the pilot with all the information he needs in a usable form and to allow him to act upon it with maximum ease and minimum reaction time.

In place of the confusing array of dials and instruments that once confronted the pilot, the modern cockpit has a head-up display (HUD) and, typically, three multifunction display (MFD) color screens. On these the pilot can call up the information he needs – a moving map overlaid with target and flight information, a range of different radar displays, information on threats identified by the aircraft’s ESM (Electronic Surveillance Measures), and so on. The HOTAS (Hands on Throttle and Stick) system pioneered by the F-15 Eagle enables the pilot to respond to a threat or engage a target without taking his hands off either the throttles or the control column. An array of switches and buttons – in the F-15, nine on the throttles and six on the stick – allows the

pilot to perform vital tasks such as designating a target, selecting and launching a weapon, and dispensing chaff or flares to distract an incoming missile. Although it requires considerable digital dexterity to operate, the HOTAS system means that a pilot in the heat of combat no longer loses vital seconds groping around for the switch he needs, just as the HUD allows him to absorb essential information without taking his eyes off the sky.

One effect of these innovations has been to make single-seat fighters fashionable again. The accepted wisdom was once that the complexity of a modern fighter aircraft’s avionics and weapons systems made it impossible for a pilot to fly alone, but improvements in cockpit design and the use of computers have greatly reduced the workload.

F-16 COCKPIT

The cockpit of an F-16 single-seat fighter shows the color multifunction visual displays, on which the pilot can call up the information he needs at the touch of a button, and the switches on the throttles and control stick, which enable the pilot to launch weapons or deploy countermeasures with his hands still on the controls.

HELMET AND MASK

The pilot’s helmet is lightweight to minimize neck injuries during combat maneuvers. The mask acts both as the intercom and source of oxygen supply.

Lightweight helmet with headphones

Oxygen mask with microphone for radio

Single lens tinted visor

Emergency jettison

Throttle

HEAD-UP DISPLAY (HUD)

A head-up display projects information into the pilot’s line of sight – an extremely useful gadget when it comes to landing an aircraft.

